Experiment 10: (Advanced Programs using linked list)

Q1) Write a program to create a single linked list for storing the N cricket player details having member's player name, team name and batting average. Display only those players information whose batting average>=50

```
#include<stdio.h>

#include<stdlib.h>

#include<string.h>

struct node{

   char name[50];

   char team[50];

   float avg;

   struct node *next;

};

struct node *head=NULL;

void create(int n){

   struct node *temp,*ptr;

   int i;

   for(i=0;i<n;i++){

      temp=(struct node*)malloc(sizeof(struct node));

      printf("Enter player name: ");

      scanf("%s",temp->name);

      printf("Enter team name: ");

      scanf("%s",temp->team);

      printf("Enter batting average: ");

      scanf("%f",&temp->avg);

      temp->next=NULL;

      if(head==NULL)

         head=temp;
```

```
        else{
            ptr=head;
            while(ptr->next!=NULL)
                ptr=ptr->next;
            ptr->next=temp;
        }
    }
}
void display(){
    struct node *temp=head;
    printf("\nPlayers with batting average >= 50\n");
    while(temp!=NULL){
        if(temp->avg>=50)
            printf("%s %s %.2f\n",temp->name,temp->team,temp->avg);
        temp=temp->next;
    }
}
int main(){
    int n;
    printf("Enter number of players: ");
    scanf("%d",&n);
    create(n);
    display();
    return 0;
}
```

Q2) Write a program to create a double linked list for storing account details of bank customers such as AC no, name, balance. Store details for N bank account holders and find the total balance for all account holders.

```
#include<stdio.h>
#include<stdlib.h>
struct node{
   int acno;
   char name[50];
   float balance;
   struct node *prev;
   struct node *next;
};
struct node *head=NULL;
void create(int n){
   struct node *temp,*ptr;
   int i;
   for(i=0;i<n;i++){
      temp=(struct node*)malloc(sizeof(struct node));
      printf("Enter account number: ");
      scanf("%d",&temp->acno);
      printf("Enter name: ");
      scanf("%s",temp->name);
      printf("Enter balance: ");
      scanf("%f",&temp->balance);
      temp->next=NULL;
      temp->prev=NULL;
      if(head==NULL)
         head=temp;
```

```
        else{
            ptr=head;
            while(ptr->next!=NULL)
                ptr=ptr->next;
            ptr->next=temp;
            temp->prev=ptr;
        }
    }
}
void total_balance(){
    struct node *temp=head;
    float total=0;
    while(temp!=NULL){
        total+=temp->balance;
        temp=temp->next;
    }
    printf("\nTotal Balance of all customers = %.2f\n",total);
}
int main(){
    int n;
    printf("Enter number of customers: ");
    scanf("%d",&n);
    create(n);
    total_balance();
    return 0;
}
```